UNIT 4: Transducers and Measurement Errors

Syllabus topics: *Functional Elements of Instruments, Classification & Characteristics, Types of Errors, Sources of Error, Dynamic Characteristics, Active and Passive Transducers: Resistive Transducers, Thermistor, Strain Gauge, Thermocouple, Differential Output Transducers, LVDT and their Characteristics.*

Question 1

4(a)	Discuss the principle of operation of a Linear Variable Differential Transformer (LVDT). Also, state its advantages and disadvantages.	5	CO4	K2, K4
4(b)	What is transducer? Explain the difference between active and passive transducer with examples.	5	CO4	K1, K2

END SEM EXAM 2024-25 | NET-101 | CE/CSE/EE/ET/IT/ME | Q4(a) Principle of operation of LVDT + advantages & disadvantages, Q4(b) Transducer — difference between active & passive transducers with examples

Question 2

4.(a)	Show the working of LVDT with proper diagrams.	5	4
(b)	Expected value of voltage across a resistor is 90 V. However measured value is 88.5 V. Calculate (i) Percentage Error (ii) Relative Accuracy	2	4
(c)	Explain the functional block diagram of an instrument.	3	4
5.(a)	Draw the working of CRO with proper diagram. Write the measurements which can be done using CRO.	6	5

END SEM PYQ's | NET101 | CS/CE/EC/EE/IT/ME | Q4(a) Working of LVDT with proper diagrams, Q4(b) Expected vs measured voltage — percentage error & relative accuracy, Q4(c) Functional block diagram of an instrument

Question 3

output and draw its characteristic table.

Q4(a)	The expected value of the current through a resistor is 20 mA. However the measurement yields a current value of 18 mA. Calculate (i) absolute error (ii) % error and (iii) relative accuracy	3	CO4	K5
Q4(b)	The below table gives the set of ten measurement that were recorded in the laboratory. Calculate the precision of 5th measurement.	2	CO4	K5

Measurement Number	Measurement Value
1.	98
2.	101
3.	102
4.	97
5.	101
6.	100
7.	103
8.	98
9.	106
10.	99

Practice / Model Question Set | NET-101 | Q4(a) Absolute error, % error & relative accuracy for measured current, Q4(b) Precision of 5th measurement from a 10-reading data table, Q4(c) Construction & working of LVDT